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Optimal Sizing of BESS for Customer Demand Management

Kyeong-hee Cho*, Seoul-ki Kim* and Eung-sang Kim†

Abstract – The paper proposes an optimal sizing method of a customer's Battery Energy Storage System(BESS) which aims at managing the electricity demand of the customer to minimize electricity cost under the time of use(TOU) pricing. Peak load limit of the customer and charging and discharging schedules of the BESS are optimized on annual basis to minimize annual electricity cost, which consists of peak load related basic cost and actual usage cost(energy cost). The optimal scheduling is used to assess the maximum cost savings for all sets of candidate capacities of BESS. An optimal size of BESS is determined from the cost reduction rate(hereinafter reduction rate) curves via capacity of BESS. Case study uses real data from industry customer and shows how the proposed method can be employed to optimally design the size of BESS for customer demand management.

Keywords: Battery energy storage system, Optimal sizing, Charging and discharging schedule, Peak load limit, Demand management, Electric cost, Reduction rate

1. Introduction

Due to the importance of demand management and the energy storage device dissemination policy, the time of use pricing that had been applied to consumers with contract demand of 1,000[kW] or more has been extended as the consumers with contract demand of 300[kW] or more and planning to be extended to the consumers of about 160,000 houses with 100[kW] or greater in the future [1, 2]. In case of the consumers getting applied with the time of use pricing, about 30~40% rate increase is expected. As the gap of time of use pricing is planning to be extended, it seems that the need of Battery Energy Storage System(BESS) for the demand management of consumers would increase rapidly.

As BESS has high investment costs, it is very important to design proper capacity by analyzing the cost saving effect followed by demand management operation precisely. In relation to this, variable researches have been performed [3~7].

This paper presents a method on optimal sizing of BESS for customer demand management in order to reduce the electricity cost under the time of use pricing. The maximum reduction rate of electricity cost by presenting a model to establish a detailed annual optimum charging and discharging schedule on the rated output and capacity

candidate group to propose a method of selecting the optimum capacity from the relationship of maximum reduction rate on the rated output and capacity gained from the calculation result. Calculating the maximum reduction rate according to the capacity is the point in selecting the optimum capacity while this depends on the comprehensiveness and accuracy on setting up optimum charging and discharging schedule of BESS. In the case study, the optimum charging and discharging schedule was set up to estimate the maximum reduction rate and showed the process of selecting the optimum capacity based on the actual data of one consumer using the proposed method.

2. Calculation of Electricity Cost

The electric charge system being applied by power companies are composed of base rate and energy rate while the amount of charge is determined by including the power industry infrastructure fund (3.7%) and VAT (10%) to the sum base rate and energy rate. Therefore, the monthly electricity cost is calculated as the formula (1) ~ (3).

$$\text{Energy Cost[Won]} = \left[\sum_{i=1}^{24} (P_i \times C_i) \right] \times n \times 1.137 \quad (1)$$

$$\text{Base Cost[Won]} = \text{Contract demand[kW]} \times \text{Demand charge[Won/kW]} \times 1.137 \quad (2)$$

$$\text{The monthly electricity cost[Won]} = \text{Energy cost[Won]} + \text{Base cost[Won]} \quad (3)$$

Here, i : Time[hour]

n : The number of days in the month

† Corresponding Author: Smart Distribution Research Center, Korea Electrotechnology Research Institute, Korea (eskim@keri.re.kr)

* Smart Distribution Research Center, Korea Electrotechnology Research Institute, Korea (kx1004xh@keri.re.kr)

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P_i : Energy in i hour[kWh]

C_i : Energy charge in i hour[Won/kWh]

The unit price of base rate and energy rate vary depending on the power supply method (high voltage or lower voltage), by contracted type (for resident, for general purposes, for industries, for education or for agriculture, etc.) while the application of base rate on all contract types except for the power for residential is based on the contract demand. However, the base rate may be estimated using peak load among peak load of December, January, February, July, August, September or this month's portion from the last 12 months including the month of checking the meter for the customers who have installed the digital power meter [8].

3. Optimal Sizing Method of BESS

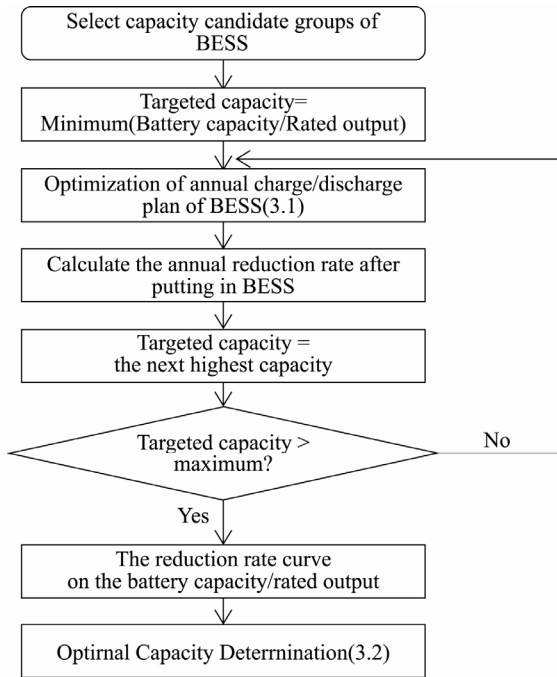


Fig. 1. Optimal Capacity Determination Method of BESS.

Fig. 1 is the overall flow chart on the optimal capacity determination method of consumers.

First, many capacity candidate groups formed by the combination of BESS battery capacity and rated output of charge/discharge device (hereinafter battery capacity/rated output) are selected. Among these, the optimum annual charging and discharging schedule is set up using the combination with minimum battery capacity/rated output as the target capacity while calculating the reduction rate after

putting in the BESS. As the same calculation is performed repeatedly by setting the next highest capacity as the target capacity, the target capacity becomes the maximum capacity combination of the capacity candidate group.

The annual maximum reduction rate according to all capacity candidate groups is calculated. The maximum reduction rate curve on the battery capacity/rated output can be drawn. Then the optimum battery capacity/rated output can be estimated based on the results.

3.1 Optimization of Annual Charging and Discharging Schedule of BESS

The lowest limit value of peak load is estimated on the battery capacity/rated output of target BESS in Figure 1 to optimize the daily charging and discharging schedule of BESS.

The demand by hour of each day according to the optimized daily charging and discharging schedule of the representative date of each month is called the load so that the monthly demand and annual demand can be calculated as formula (4) ~ (5) by multiplying the number of days by each month while the annual electric cost can be calculated as formula (6) using formula (1) ~ (3).

$$\text{The monthly demand [kWh]} = \left(\sum_{i=1}^{24} P_i [\text{kW}] \right) \times n \quad (4)$$

$$\text{The annual demand [kWh]} = \left[\sum_{j=1}^{12} \sum_{i=1}^{24} P_i [\text{kW}] \right] \times n_j \quad (5)$$

$$\text{The annual electric cost [Won]} = \sum_{j=1}^{12} \sum_{i=1}^{24} \left(\begin{aligned} &P_{ij} [\text{kWh}] \times C_{ij} [\text{Won / kWh}] \times n_j \\ &+ \text{Contract demand [kW]} \\ &\times \text{Demand charge [Won / kW]} \end{aligned} \right) \quad (6)$$

Here, j : The number of month

n_i : The number of days in j month

P_i : Energy in i hour in j month [kWh]

C_i : Energy charge in i hour [Won/kWh]

Perform same operation until the lowest limit of peak load becomes identical as the maximum load of the consumer before installing BESS by repeating the optimization of daily charging and discharging schedule as the annual electric cost is calculated and stored to consistently increasing the lowest limit of peak load in Fig. 2. If the annual electric cost according to the increase of lowest limit of peak load is compared, while the base rate gets reduced as the maximum load gets lower, the energy rate increases or converges as the charge/discharge range of battery gets more restriction. Therefore, there is a nature of being in conflict by the maximum load although the

reduction of base rate and reduction of energy rate have difference in their degree. The minimum annual electric cost is discriminated from the result of annual electric cost on the various setup of maximum load gained by repeated performance and this is selected as the optimum annual charging and discharging schedule.

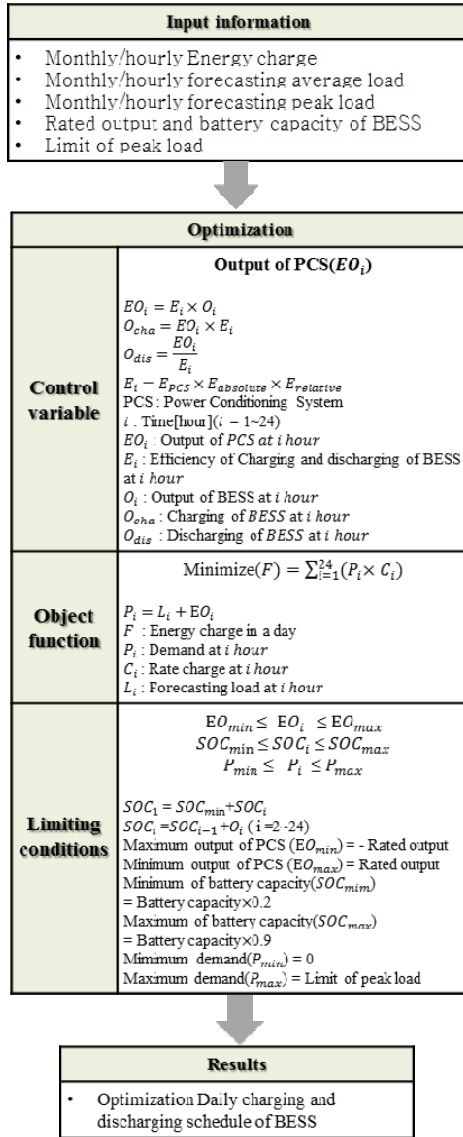


Fig. 2. Optimization Daily Charging and Discharging Schedule of BESS.

3.1.1. Calculating the Lowest Limit of Peak Load

The method of estimating the lowest limit of peak load to reduce the peak load as much as possible while satisfying the restricting conditions using BESS is followed.

First, the value that has subtracted the rated output of

storage system from the peak load of consumer is set as the initial value on the lowest limit of peak load and performed the optimizing operation on the maximum load date of each month from January to December. If the solution does not exist, calculate the lowest limit by repeating performance until the optimum solution exists by increasing the lowest limit. Thus, the base rate can be reduced as much as possible by limiting the peak load using the lowest limit of peak load.

3.1.2. Optimization Daily Charging and Discharging Schedule

The process on the optimization of daily charging and discharging schedule of BESS can be shown as Fig. 2.

In order to create the optimization model, four factors including the input information of model, control variable, object function and limiting conditions must be entered. This paper proposes the optimization function to build up the daily charging and discharging schedule of the storage system with the minimized energy rate as object function by setting up the output of PCS as the control variable. A nonlinear optimization method was used to optimize by setting the minimization of annual electric cost by multiplying the monthly number of days to the daily energy rate according to the daily charging and discharging schedule of the storage system as final object function.

Efficiency of charging and discharging of BESS of control variable can be calculated that the commonly used PCS efficiency was entered as 95% and the absolute efficiency was entered as 100% here.

3.2 Optimal Capacity Determination

The process of estimating the maximum capacity of BESS using the maximum reduction rate result of the annual electric cost by each capacity candidate group of BESS is followed.

First of all, calculate the reduction rate curve on the candidates rated output of BESS and its base capacity combination. Here, the base capacity is defined by the battery capacity that has multiplied the parameter to the rated output. In the reduction rate curve on the candidates rated output, the interval with the greatest slant is calculated. The optimum rated output is estimated from this interval, then the rate reduction rate curve by varying the battery capacity is illustrated, the interval with greatest slant is found by calculating the slant of reduction rate on the battery capacity and the greatest value among these intervals is selected as the optimum battery capacity.

4. Case Study

4.1 Analysis on the Load Pattern of Target Consumers

The monthly load pattern of business day, legal holiday and peak day of the target consumer was shown as Fig. 3~5.

The target consumer of this paper is one of industry, the contracted power is 8,500[kW] and the peak load of surveyed load data in 2012 was 38,707.2[kW].

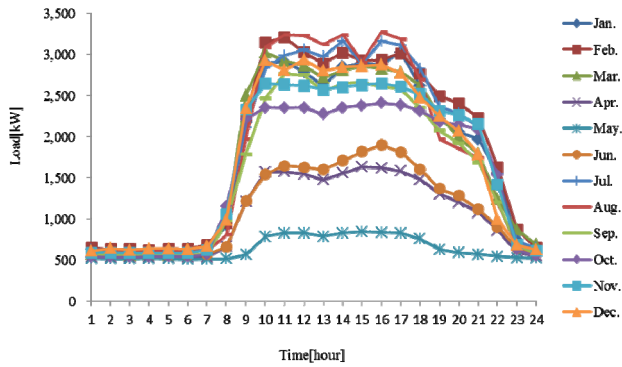


Fig. 3. Monthly Load pattern curve during business days.

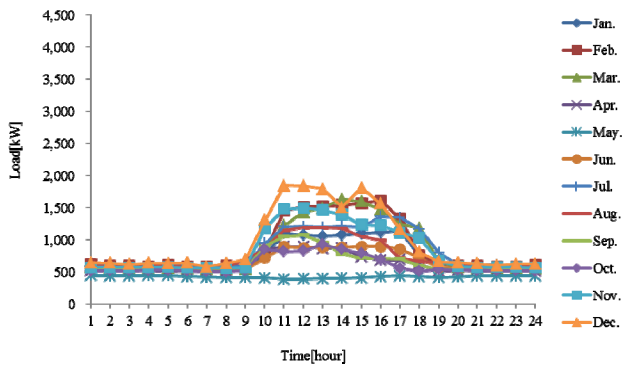


Fig. 4. Monthly Load pattern curve during legal holidays.

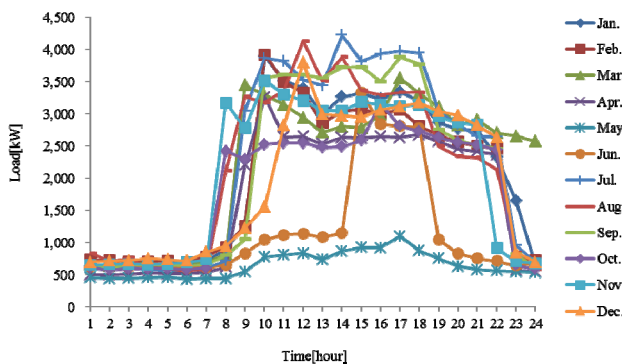


Fig. 5. Monthly Load pattern curve during a peak day.

The monthly one day load pattern from January to December has organized the survey data of target consumers as monthly and day of week to be classified based on business day and legal holiday to be averaged by each hour.

Using the monthly peak load date on the survey load data of the target consumer, the pattern of monthly peak load date was calculated as shown in Fig. 5.

4.2 Optimization of Charging and Discharging Schedule of BESS

The standard electricity cost before putting in BESS of 2012 has been calculated using formula (1) ~ (3) after multiplying respectively load data of Fig. 3~5 and a number of days of each month.

Table 1. The Reduction Rate according to Case #

Case#	Rated output [kW]	Battery capacity [kWh]	Limit of peak load [kW]	Reduction rate [%]
1	400	3,200	3,735.2	6.4
2	600	4,800	3,535.2	9.3
3	800	6,400	3,335.2	12.1
4	1,000	8,000	3,135.2	14.8
5	1,200	9,600	2,935.2	17.2
6	1,400	11,200	2,744.4	19.2
7	1,600	12,800	2,638.9	20.9
8	1,800	14,400	2,533.6	22.4
9	2,000	16,000	2,435.0	23.8
10	2,200	17,600	2,375.4	25.1
11	2,400	19,200	2,374.6	25.9
12	2,600	20,800	2,374.6	25.9
13	2,800	22,400	2,374.6	26.1
14	3,000	24,000	2,374.6	26.2

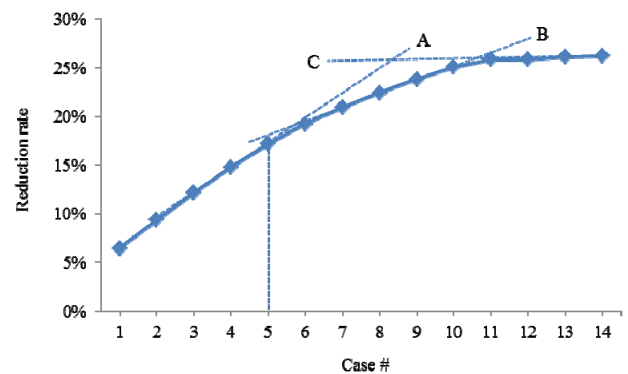


Fig. 6. The Reduction Rate Curve according to case of BESS.

The capacity candidate group of BESS on the target consumer has fixed the rated output range as 400~3,000[kW] and the parameter of the battery capacity as 8. The annual charging and discharging schedule of BESS

has been optimized to calculate the electric cost using the algorithm of optimum capacity selection method of BESS mentioned before by increasing the rated output 200[kW] at a time. The result of rate reduction rate according to the capacity candidate group of BESS in Table 1 has been shown as the graph of Fig. 6.

The reduction rate curve in Fig. 6 can be divided into the interval with slant of A, B and C. As slant A is the greatest, the interval which is the rated output 400~1,200[kW] becomes the optimum rated output range with the greatest reduction rate effect. Therefore, as the highest efficiency compared to the capacity is produced when the output of BESS is 1,200[kW] and the capacity is 9,600[kWh], 1,200[kW] could be selected as the optimum rated output.

The annual charging and discharging schedule has been optimized to show the result as table 2 by fixing the optimum rated output as 1,200[kW] according to chapter 3.2 and changing the battery capacity as 1~15 time rate.

The rate reduction rate graph on each capacity of table 2 was shown as Fig. 7.

Table 2. The Reduction Rate according to Case #

Case#	Rated output [kW]	Battery capacity [kWh]	Limit of peak load [kW]	Reduction rate [%]
1	1,200	1,200	3,653.0	4.7
2	1,200	2,400	3,411.0	6.6
3	1,200	3,600	3,263.5	9.6
4	1,200	4,800	3,172.4	11.5
5	1,200	6,000	3,090.9	13.1
6	1,200	7,200	3,010.1	14.7
7	1,200	8,400	2,935.2	16.2
8	1,200	9,600	2,935.2	17.2
9	1,200	10,800	2,935.2	18.1
10	1,200	12,000	2,935.2	18.7
11	1,200	13,200	2,935.2	19.0
12	1,200	14,400	2,935.2	19.0
13	1,200	15,600	2,935.2	18.9
14	1,200	16,800	2,935.2	18.9
15	1,200	18,000	2,935.2	19.0

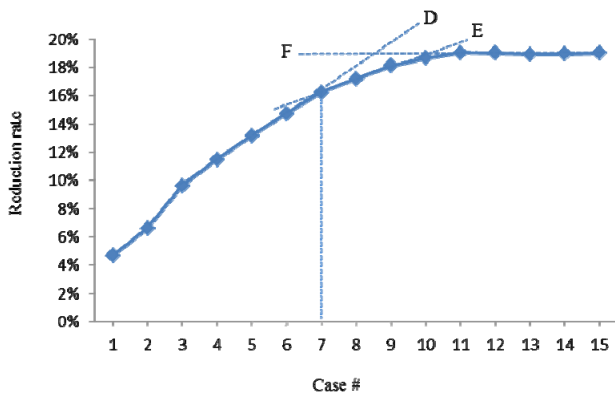


Fig. 7. The Reduction Rate Curve according to Cases.

In Fig. 7, the reduction rate curve can be divided into the interval with slant of D, E and F. As slant D is the greatest, case 7 becomes the optimum battery capacity. Therefore, it is reasonable to select the rated output of 1,200[kW] and battery capacity of 8,400[kWh] as the optimum capacity of BESS.

4.3 Results using Optimal Capacity BESS

Through the case study, the BESS with rated output of 1,200[kW] and battery capacity of 8,400[kWh] could be selected as the optimum capacity of target consumer.

According to charging and discharging schedule of optimal capacity BESS, the fact that load graph (before putting in BESS) gets changed as demand graph (after putting in BESS) and the demand graph gets restricted according to the demand limit could be seen.

The charging and discharging scheduling of BESS is shown as Fig. 9. This is the optimum charging and discharging schedule of BESS compared to the capacity that has optimized both base rate and energy rate while the result on the peak load has been shown in Fig. 9.

The change of electric cost followed according to BESS input is shown as Fig. 10.

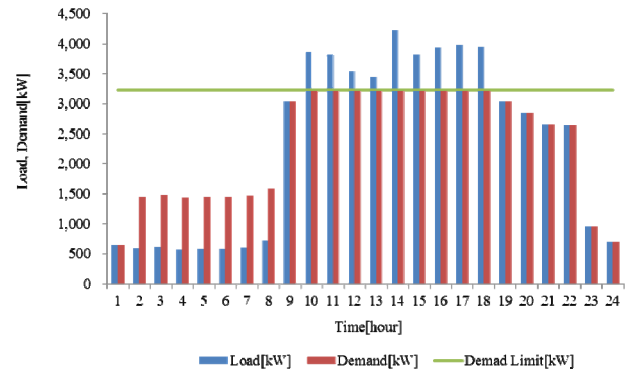


Fig. 8. Load and Power during a Peak Day.

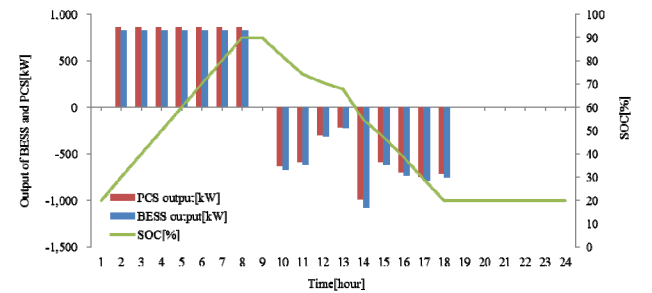


Fig. 9. BESS and PCS Output during a Peak Day.

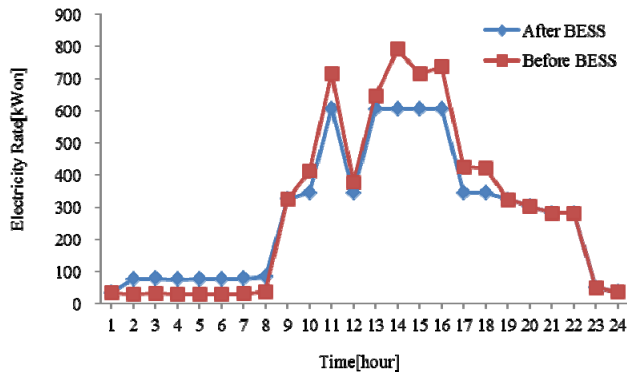


Fig. 10. Change of the Energy Cost by Optimal BESS under the Time of Use Pricing during a Peak Day.

After the BESS input, the load gets reduced by the discharge of peak time slot and the load by the charge of off-peak time slot has increased so that the reduction on the area of graph which is the total rate according to time-of-use pricing could be seen.

5. Conclusion

In this paper we proposed the method of optimal sizing of Battery Energy Storage System (BESS) for consumer demand management. In order to gain the annual maximum reduction rate, the established model on the annual charging and discharging schedule of BESS that has optimized the annual peak load limit value and hourly charging and discharging schedule of battery system have been presented to suggest the method of selecting the optimum capacity based on the maximum reduction rate of electric cost toward the rated output and battery capacity derived from these.

This study was focused on presenting a method of optimizing BESS that can be used on the site by adequately reflecting the characteristics of consumer load, considering the efficiency of battery according to the charge/discharge speed as well as presenting a model to optimize the maximum demand limit value and hourly charging and discharging schedule of the consumer having direct impact on base rate and the energy rate after analyzing the load pattern of monthly business days, legal holidays and peak load days of the target consumer in detail. A design method that can be more generally used is planning to be developed through the development of an algorithm that has adjusted to the economic efficiency model including future investment costs.

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Kyeong-Hee Cho received M.S degree in electrical engineering from Gyeongsang National University. Currently, she works as a researcher in the smart distribution research center at KERI. Her research interests are Smart distribution, operation and reliability of power grid.



Seul-Ki Kim received his B.S., M.S and Ph.D. degrees in Electrical Engineering from Korea University. Currently, he works as a senior researcher in the smart distribution research center at KERI. His research interests are in grid interface of distributed generators and modeling and analysis of distributed generation resources.



Eung-Sang Kim received his B.S. degree in Electrical Engineering from Seoul National University of technology and his M.S. and Ph.D. degree in electric engineering from Soong-sil University. Currently, he works as a director in the smart distribution research center at KERI. His research interests are in the areas of power quality, dispersed generating system integration and application, and grid-connection of dispersed generations.